

THE BIG BANG

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TIME, SPACE, ENERGY, AND MATTER are all thought to have come into existence 13.8 billion years ago, in the event called the Big Bang. In its first moments, the universe was infinitely dense, unimaginably hot, and contained pure energy. But within a tiny fraction of a second, vast numbers of elementary particles had appeared, created out of energy as the universe cooled. Within a few hundred thousand years, these particles had combined to form the first atoms.

IN THE BEGINNING

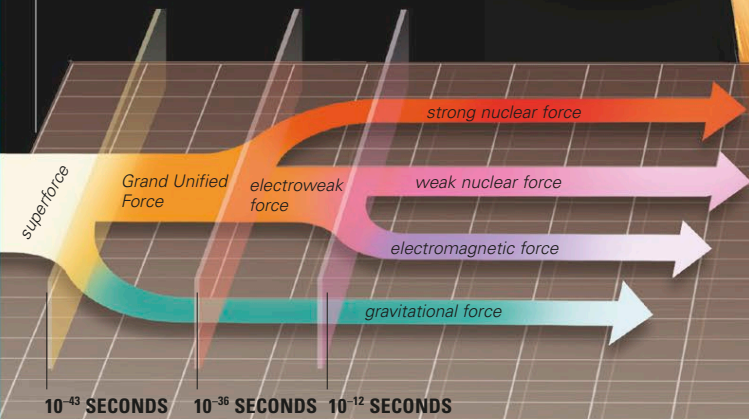
The Big Bang was not an explosion in space, but an expansion of space, which happened everywhere. Physicists do not know what happened in the first instant after the Big Bang, known as the Planck epoch, but at the end of this period, they believe that gravity split from the other forces of nature, followed by the strong nuclear force (see p.30). Many believe this event triggered “inflation”—a short but rapid expansion. If inflation did occur, it helps to explain why the universe seems so smooth and flat. During inflation, a fantastic amount of mass-energy came into existence in tandem with an equal but negative amount of gravitational energy. By the end of inflation, matter had begun to appear.

THE PLANCK EPOCH
No current theory of physics can describe what happened in the universe during this time.

singularity at the start of time

GRAND UNIFIED THEORY EPOCH

During this epoch, matter and energy were completely interchangeable. Three of the fundamental forces of nature were still unified.

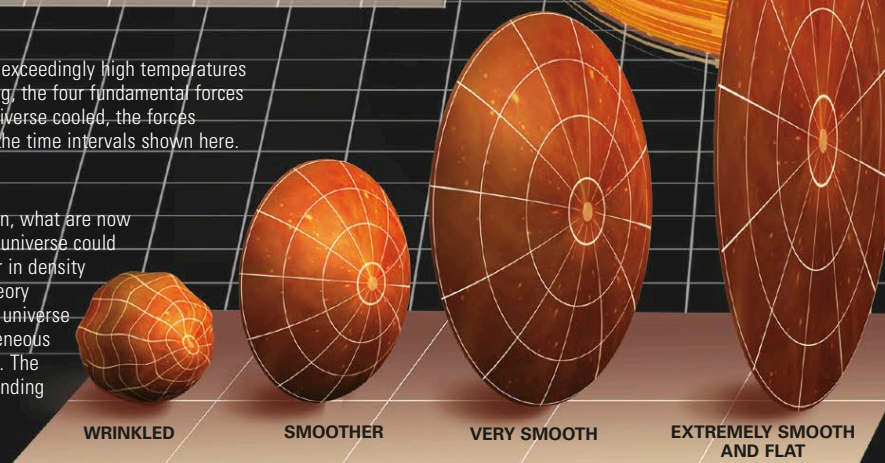


SEPARATION OF FORCES

Physicists believe that at the exceedingly high temperatures present just after the Big Bang, the four fundamental forces were unified. Then, as the universe cooled, the forces separated, or “froze out,” at the time intervals shown here.

INFLATION

In a Big Bang without inflation, what are now widely spaced regions of the universe could never have become so similar in density and temperature. Inflation theory proposes that our observable universe is derived from a tiny homogeneous patch of the original universe. The effect of inflation is like expanding a wrinkled sphere—after the expansion, its surface appears smooth and flat.



THE FIRST MICROSECOND

The timeline on this page and the next shows some events during the first microsecond (1 millionth of a second or 10⁻⁶ seconds) after the Big Bang. Over this period, the universe's temperature dropped from about 10²⁶°C (10 billion trillion trillion degrees) to a mere 10¹¹°C (100 billion degrees). The timeline refers to the diameter of the observable universe: this is the approximate historical diameter of the part of the universe we can currently observe.

DIAMETER	3x10 ⁻²⁶ ft/10 ⁻²⁶ m	33 ft/10 m	10 ⁵ m (62 miles/100 km)
TEMPERATURE	10 ²⁸ K (18,000 trillion trillion °F/10,000 trillion trillion °C)		10 ²² K (18 billion trillion °F/10
TIME	A thousand-billionth of a yoctosecond 10 ⁻³⁶ seconds	A hundred-millionth of a yoctosecond 10 ⁻³² seconds	1 yoctosecond 10 ⁻²⁴ seconds

INFLATION EPOCH

Part of the universe expanded from billions of times smaller than a proton to something between the size of a marble and a football field.

ELECTROWEAK EPOCH

This period, which followed separation out of the strong nuclear force, saw vast numbers of quark and antiquark pairs forming from energy and annihilating back to energy. Gluons and other more exotic particles also appeared.

a ten-trillionth of a yoctosecond
10⁻⁴³ seconds



PARTICLE SOUP

About 10⁻³² seconds after the Big Bang, the universe is thought to have been a “soup” of elementary particles and antiparticles. These were continually formed from energy as particle-antiparticle pairs, which then met and were annihilated back to energy. Among these particles were some that still exist today as constituents of matter or as force-carrier particles. These include quarks and their antiparticles (antiquarks) and bosons such as gluons (see pp.30–31). Other particles that are hard to detect today will also have been present. These include Higgs bosons, which impart mass to other particles, and possibly gravitons (hypothetical gravity-carrying particles).